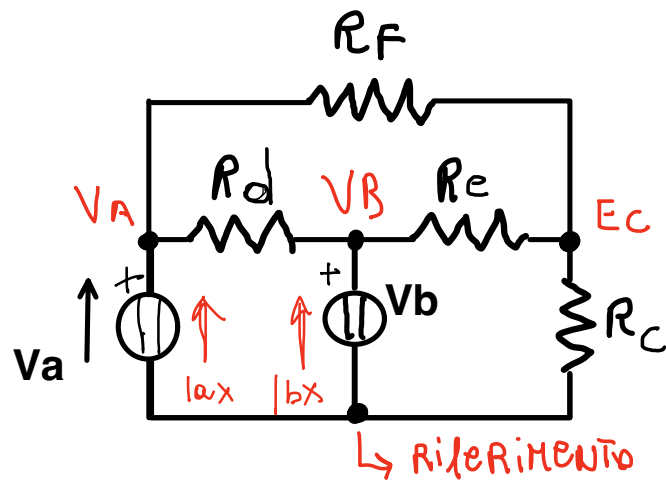




$$V_a = 15^0; R_C = 1^0; R_e = 2^0;$$

$$V_b = 1^0; R_d = 1^0; R_f = 3^0;$$

Si DETERMINANO tutte le tensioni e le correnti sui vari componenti e si effettui la verifica del Bilancio delle potenze.

METODO DEI NODI.

$$\begin{bmatrix} G_F + G_d & -G_d & -G_F \\ -G_d & G_d + G_e & -G_e \\ -G_F & -G_e & G_F + G_e + G_C \end{bmatrix} \begin{pmatrix} V_A \\ V_B \\ E_C \end{pmatrix} = \begin{pmatrix} I_{aX} \\ I_{bX} \\ 0 \end{pmatrix}$$

$$\begin{cases} (G_F + G_d) V_A - G_d V_B - G_F E_C = I_{aX} \\ -G_d V_A + (G_d + G_e) V_B - G_e E_C = I_{bX} \\ -G_F V_A - G_e V_B + (G_F + G_e + G_C) E_C = 0 \end{cases}$$

$$\left\{ \begin{array}{l} \frac{4}{3} 15 - 1 - \frac{1}{3} E_C = I_{aX} \\ -15 + \frac{3}{2} - \frac{1}{2} E_C = I_{bX} \\ -\frac{1}{3} 15 - \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{2} + \frac{1}{1} \right) E_C = 0 \end{array} \right.$$

$$\left\{ \begin{array}{l} 19 - \frac{1}{3} E_C = I_{aX} \\ -\frac{27}{2} - \frac{1}{2} E_C = I_{bX} \\ -\frac{11}{2} + \frac{11}{6} E_C = 0 \end{array} \right. \rightarrow \left\{ \begin{array}{l} I_{aX} = 18 \checkmark \\ I_{bX} = -15 \checkmark \\ E_C = 3 \checkmark \end{array} \right.$$

BILANCIO DI POTENZA

$$V_{Rd} = V_B - V_A = 1 - 15 = -14$$

se considero



MA se considero



$$V_{Rd} = V_a - V_b = 15 - 1 = 14$$

$$V_{Rf} = E_C - V_A = 3 - 15 = -12$$

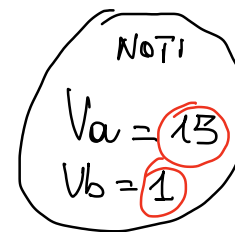
se considero



MA se considero



$$V_{Rf} = 15 - 3 = 12$$



$$E_C = 3$$

$$V_{Re} = E_C - V_A = 3 - 1 = 2 \quad V_{RC} = E_C = 3$$

$$I_{Rd} = \frac{V_{Rd}}{R_d} = \frac{14}{1} = 14 \quad I_{Rf} = \frac{V_{Rf}}{R_f} = \frac{12}{3} = 4 \quad I_{Re} = \frac{V_{Re}}{R_e} = \frac{2}{2} = 1$$

$$I_{Rc} = \frac{V_{RC}}{R_c} = \frac{3}{1} = 3 \quad I_{Ax} = 18 \quad I_{bx} = -15$$

$$P_a = V_A \cdot I_A = 15 \cdot 18 = 270 \quad P_b = V_b \cdot I_b = 1 \cdot (-15) = -15$$

$$P_c = \underset{\substack{\uparrow \\ E_C}}{V_C} \cdot I_C = 3 \cdot 3 = 9 \quad P_d = V_d \cdot I_d = 14 \cdot 14 = 196$$

$$P_e = V_e \cdot I_e = 2 \cdot 1 = 2 \quad P_f = V_f \cdot I_f = 12 \cdot 4 = 48$$

$$\left. \begin{array}{l} \text{POTENZA EROGATA DAI GENERATORI} = 270 - 15 = 255 \\ \text{POTENZA EROGATA DAI RESISTORI} = 196 + 9 + 2 + 48 = 255 \end{array} \right\} 255 = 255 \quad \checkmark$$

